

# Experimental Setup

Can upstream surface data constrain separate local first/last markers, and does causal two-site capture remove the initial-data inverse sensitivity before another nonlinear reception run?

Experiment: gross.reconstruction.v1    Model: signal-space.ss-ocf-1.flat-neutral-clock.v1  
Runtime configuration: fixtures/research/gross-test-07-reconstruction.json  
Plan lock: f517a3471570bcd3cf620d77bc5d8d97e5d6037684c43ff072e475b56fda3cba

## MODEL AND NUMERICAL SETUP

Action: Frozen linear neutral sector of unchanged SS OCF 1:  $M \mathbf{b}_{tt} = -K \mathbf{b}$ , both  $M$  and  $K$   $Z$ -weighted. No nonlinear receiver.

Units:  $c = \hbar = m = 1$ ;  $\mathbf{b} = r \cdot \mathbf{a}$ ; time in  $m^{-1}$ .

Domain: Frozen independently solved  $\omega_Q = .9$  profiles from verified transfer run;  $h = .05, .025, .0125$ ,  $R = 80$ ; independent  $R100$  at  $h = .025$ .

Boundaries: Regular origin and outer Dirichlet; duration 60, local  $r = .1$ , acquisition  $r14$  to  $t16$ .

Initial Data: Support[18,26],  $\mathbf{v} = D_h \mathbf{b}$ ; broad and carrier prior shapes plus new center 21.5 width 2.75 carrier.8. All shapes fixed before execution.  $A = .012$ .

Solver: RK4  $dt = .0025$  and finest  $.00125$ . Exact adjoint  $H$  and  $L$ ; full economy SVD. Two-site causal commutator forcing from  $\mathbf{b}$  and  $\mathbf{b}_t$  every  $.0125$ , stopped at  $t16$ ; sample  $.025$  control.

Tolerances: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta = 1e-10 ||\text{surface}||$ ,  $\rho = 2 ||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## COMPETING PREDICTIONS

observable: Surface-consistent variations constrain both events to the declared target; causal capture gives controlled local markers.

ill-conditioned: Weak singular directions move late events despite tiny surface residual, or finite acquisition/sampling leaves causal timing unresolved.

## CONTROLS

numeric: Separate each source of numerical error; method: Three meshes, independent finest time halving, wider box, independent surface/output decimation.

singular: Test interior sensitivity rather than surface fit; method: Retained/discarded decomposition, explicit input norm prior and surface tolerance, finite adversarial witnesses.

causal: Avoid full initial-state inversion; method: Two-site commutator source reconstructs projected interior state at  $t16$ ; omitted exterior state remains an audited error.

spectrum: Avoid judging only previously inspected shapes; method: All methods locked on three linear shapes before their interior audits; new shape is only a linear validation, not held-out nonlinear reception.

## PREREGISTERED DECISIONS

# Experimental Setup

## `prediction-first (prediction-first)`

Pass: Each variant locks both reconstruction forecasts and sensitivities before its known-source interior audit; every frozen input and forecast hash verifies.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.

Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.

Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## `decomposition (decomposition)`

Pass: Every source-minus-inverse local history equals the separately propagated retained plus discarded input error to relative  $1e-8$ , with absolute floor  $1e-10$ .

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.

Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.

Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## `surface-replay (surface-replay)`

Pass: All synthetic single-site inverse replay residuals are below  $1e-7$ . This check alone does not establish interior observability.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.

Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.

Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## `inverse-observability (inverse-observability)`

Pass: Each finest event conditional first-order timing bound is  $\leq 1e-4$  and no admissible finite singular-direction witness exceeds  $1e-4$ . A witness beyond target rejects robustness; a bound above target without witness is unresolved.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.

Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.

Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## `causal-capture (causal-capture)`

Pass: Every variant causal capture state differs from the projected acquired state by  $< 1e-6$  in relative quadratic energy. This excludes omitted exterior energy, which is reported separately.

# Experimental Setup

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.  
Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.  
Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

causal-timing (causal-timing)

Pass: For all three spectra each finest causal first/last timing residual is  $\leq 1e-4$  and the separate residual budget B (mesh residual change + time residual change + domain residual change + surface/output sampling  $+1e-7$ ) is  $\leq 1e-4$ ; require residual  $\leq B$ . Missing markers or  $B > \text{target}$  is unresolved, resolved violation fails.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.  
Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.  
Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

continuum-events (continuum-events)

Pass: For both methods and source reference, each finest absolute event mesh change is  $\leq 0.6$  preceding change plus time/domain/sampling differences and  $1e-7$ ; otherwise unresolved. No absolute continuum timing accuracy is implied.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.  
Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.  
Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

sampling (sampling)

Pass: Each finest event changes  $\leq 1e-4$  under output decimation .0125 to .025; causal surface sampling .0125 to .025 is separately  $\leq 1e-4$ . Otherwise unresolved.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.  
Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.  
Uncertainty: Separate first/last target  $1e-4$ , floor  $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

source-energy (source-energy)

Pass: Frozen unforced acquisition conserves its quadratic neutral energy to relative  $1e-6$  over  $t=0..16$  on every grid and waveform. No clock radiation is inferred.

Fail: A resolved violation of the specified gate; see protocol for conditional witness rules.

# Experimental Setup

Unresolved: Missing events, inadequate budget or unconverged sequence; never replace uncertainty by pass.  
Uncertainty: Separate first/last target $1e-4$ , floor $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

## VISUAL QUESTIONS

local-reconstruction [evidentiary]: How do the inverse and causal surface capture reproduce the local waveform and its late tail?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface record, local neutral history, separate marker times, singular sensitivities, direct numerical differences

observable: controlled event bounds and causal replay

ill-conditioned: surface-compatible marker shifts or unresolved timing

Controls: numeric, singular, causal, spectrum; uncertainty: Separate first/last target $1e-4$ , floor $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/singular.csv; transforms: Hermite event roots; SVD projections; unextrapolated differences; no fitted correction.

singular-observability [evidentiary]: Which surface singular directions remain relevant to each local event?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface record, local neutral history, separate marker times, singular sensitivities, direct numerical differences

observable: controlled event bounds and causal replay

ill-conditioned: surface-compatible marker shifts or unresolved timing

Controls: numeric, singular, causal, spectrum; uncertainty: Separate first/last target $1e-4$ , floor $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/singular.csv; transforms: Hermite event roots; SVD projections; unextrapolated differences; no fitted correction.

event-budgets [evidentiary]: Are first and last reconstruction errors separately below their uncertainty budgets?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface record, local neutral history, separate marker times, singular sensitivities, direct numerical differences

observable: controlled event bounds and causal replay

ill-conditioned: surface-compatible marker shifts or unresolved timing

Controls: numeric, singular, causal, spectrum; uncertainty: Separate first/last target $1e-4$ , floor $1e-7$ .  $\eta=1e-10||\text{surface}||$ ,  $\rho=2||\text{recovered input}||$  are explicit conditional perturbation radii. Causal timing

# Experimental Setup

budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/singular.csv; transforms: Hermite event roots; SVD projections; unextrapolated differences; no fitted correction.

sampling-continuum [evidentiary]: How much do sampling and mesh changes move each event?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface record, local neutral history, separate marker times, singular sensitivities, direct numerical differences

observable: controlled event bounds and causal replay

ill-conditioned: surface-compatible marker shifts or unresolved timing

Controls: numeric, singular, causal, spectrum; uncertainty: Separate first/last targetle-4, floorle-7. eta=1e-10||surface||, rho=2||recovered input|| are explicit conditional perturbation radii. Causal timing budgets use residual mesh/time/domain changes and both output/surface sampling; absolute event mesh changes are reported separately.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/singular.csv; transforms: Hermite event roots; SVD projections; unextrapolated differences; no fitted correction.

## RESOURCE LIMITS

```
{"max_memory_mb": 2048, "max_output_mb": 300, "max_wall_seconds": 4500}
```

## PROHIBITED INFERENCES

No nonlinear reception or clock phase prediction is executed. A pass is a reconstruction prerequisite only.

Original Test7 fail and Test8 block remain; a new held-out nonlinear forecast and separate acceptance review are required.

Two-site capture changes acquisition; do not attribute its information to the original single-site record.

Conditional linear timing sensitivity is not a rigorous nonlinear event bound.

No two-object, invariance, gravity, angular stability, topology or emergent-spacetime claim.